

DCPO: DYNAMIC CLIPPING POLICY OPTIMIZATION

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ABSTRACT

Reinforcement Learning from Verifiable Rewards (RLVR) is an effective route to improving the reasoning ability of large language models, but current GRPO-style methods leave two important learning signals underused: fixed token-level clipping can suppress rare-token exploration, and stepwise reward standardization can turn identical-reward responses into zero-gradient samples. We present **Dynamic Clipping Policy Optimization (DCPO)**, whose contribution is to make both token updates and response utilization adaptive. DCPO combines (i) dynamic clipping bounds determined by token-specific prior probabilities, which enlarge the update region for low-probability tokens while retaining clipping control, and (ii) smooth advantage standardization over cumulative training steps, which recovers useful response-level gradients when within-step rewards are identical. Across four mathematical reasoning benchmarks and four model scales, DCPO delivers stronger performance while using generated data more effectively. On Qwen2.5-Math-7B, DCPO reaches 46.7 Avg@1 and 38.8 Avg@32 on AIME24, exceeding DAPO (36.7/31.6), GRPO (36.7/32.1), and GSPO (40.0/34.9). On Qwen2.5-14B, DCPO obtains 23.3/19.0 on AIME25, compared with GRPO (13.3/10.5), DAPO (20.0/15.3), and GSPO (16.7/9.9). Beyond accuracy, DCPO raises the nonzero-advantage ratio by an average of 28% over GRPO, doubles training efficiency relative to DAPO, and reduces token clipping by an order of magnitude compared with GRPO and DAPO. These results support the paper’s central claim: explicitly adaptive clipping and cumulative advantage smoothing improve RLVR by preserving more useful token- and response-level learning signal.

1 INTRODUCTION

Reasoning-oriented large language models (LLMs) increasingly rely on reinforcement learning signals that are easy to verify and hard to imitate with supervised data alone (OpenAI, 2025; Guo et al., 2025; Comanici et al., 2025). Reinforcement Learning from Verifiable Rewards (RLVR) is central to this setting because it optimizes models with rule-based outcome rewards, making it especially useful for domains such as mathematics and coding where final answers can be checked reliably (DeepSeek-AI et al., 2024). The contribution question we address is therefore not whether RLVR is useful, but how to make its policy updates use generated reasoning data more effectively.

The GRPO family of algorithms (DeepSeek-AI et al., 2024; Liu et al., 2025; Yuan et al., 2025; Zheng et al., 2025) provides a strong RLVR baseline, but its update structure exposes two bottlenecks that directly limit data use. First, advantage values can decay to zero in later stages, producing entropy collapse and preventing sampled responses from contributing gradients. Second, fixed probability-ratio clipping can remove learning signal from rarely sampled tokens, narrowing the diversity of policy exploration precisely where reasoning-relevant alternatives may appear.

DAPO responds to these issues with Clip-Higher, which relaxes the upper clipping bound, and Dynamic Sampling, which filters uniform-reward samples so that a batch contains valid gradients (Yu et al., 2025). This prior solution clarifies the problem but also motivates our contribution: (i) **response-level inefficiency**, because regenerating or filtering samples lowers sampling efficiency and slows convergence; and (ii) **token-level clipping mismatch**, because our analysis shows that the appropriate clipping range varies by token probability rather than by a single uniform upper bound. Other approaches may avoid discarding zero-gradient responses directly, but these two sources of lost learning signal remain central.

We introduce **Dynamic Clipping Policy Optimization (DCPO)** to make these contributions explicit in the optimizer. **Contribution 1: cumulative smooth advantage standardization.** DCPO standardizes advantages using both the current responses and the cumulative reward distribution for the same prompt, so identical rewards within one step need not erase all response-level gradients. This design improves response-level exploration and data utilization (see Section 2). **Contribution 2: dynamic-adaptive token clipping.** DCPO sets clipping bounds from old token-specific probabilities instead of using only fixed upper and lower constants, giving lower-prior-probability tokens wider update ranges and therefore more room for token-level exploration (see Section 2 and Appendix A.6). **Contribution 3: response-preserving loss aggregation.** DCPO computes the mean loss over tokens within each response rather than across the full batch, preserving the relative advantage structure among responses to the same prompt and avoiding batch-level dilution.

We evaluate these contributions with a merged training set built from Math-DAPO-17k (Yu et al., 2025) and levels 3-5 of the MATH dataset (Liu et al., 2025), four model scales, and four standard mathematical reasoning benchmarks. Against GRPO, DAPO, and GSPO where available, DCPO consistently provides stronger or comparable results, with the clearest gains on the challenging AIME benchmarks: on AIME24-Avg@32, DCPO-7B scores 38.8 versus 32.1 (GRPO), 31.6 (DAPO), and 34.9 (GSPO); on AIME25-Avg@32, DCPO-14B reaches 19.0 versus 10.5 (GRPO), 15.3 (DAPO), and 9.9 (GSPO). The mechanism-level measurements align with the contribution claims: DCPO improves the average nonzero advantage ratio by 28% over GRPO, doubles training efficiency relative to DAPO, and reduces token clipping by an order of magnitude while maintaining superior performance.

Appendix A.2 provides the preliminary definitions and baseline loss and advantage calculations for GRPO, DAPO, and GSPO, so the main text can focus on where DCPO changes the optimization structure.

2 DYNAMIC-ADAPTIVE CLIPPING BOUNDS FOR RL(DAC)

The first DCPO component targets token-level exploration by replacing fixed ratio clipping with probability-dependent clipping. Probability-ratio clipping is widely used in reinforcement learning because it stabilizes policy updates and prevents large parameter shifts, especially when importance sampling is used in off-policy settings. If importance sampling reweights by the ratio $r(x)$ between the new probability $p(x)$ and the old probability $q(x)$, the estimate of $\mathbb{E}_{q(x)}[f(x)\frac{p(x)}{q(x)}]$ is unbiased, but its variance can grow substantially, as shown in Equation (1).

$$\text{Var}_{x \sim q} \left[f(x) \frac{p(x)}{q(x)} \right] - \text{Var}_{x \sim p} [f(x)] = \mathbb{E}_{x \sim p} \left[f(x)^2 \left(\frac{p(x)}{q(x)} - 1 \right) \right] = \int f(x)^2 \left(\frac{p(x)}{q(x)} - 1 \right) p(x) dx \quad (1)$$

Prior work therefore controls variance by constraining the probability ratio with a fixed hyperparameter ϵ (e.g., Schulman et al. (2017)), as written in Equation (2).

$$|r(x) - 1| \leq \epsilon \quad (2)$$

This fixed-bound design is precisely the limitation DAC is intended to address. A symmetric bound constrains exploration most strongly where the old policy assigns low probability mass, because the allowable absolute change for low-probability tokens is inherently small. Asymmetric bounds such as those used by DAPO relax part of this constraint, but they still apply a token-independent rule. GSPO instead clips at the sequence level and discards entire responses when the sequence-level importance ratio exceeds the asymmetric bounds, which can remove informative tokens while leaving other high-variance tokens to affect training stability.

DAC’s contribution is to control the variance-bias trade-off with a probability-dependent constraint rather than a globally fixed ratio interval. We formalize this constraint in Equation (3).

$$|(r(x) - 1)p(x)| \leq \epsilon \quad (3)$$

Substituting $p(x) = r(x)q(x)$ into Equation (3) and enforcing non-negativity of the square-root term with a maximum against zero gives a closed-form interval for $r(x)$. The resulting **dynamic-adaptive**

$$0.5 + \frac{1}{2} \sqrt{\max \left(1 - \frac{4\epsilon_{low}}{q(x)}, 0 \right)} \leq r(x) \leq 0.5 + \frac{1}{2} \sqrt{1 + \frac{4\epsilon_{high}}{q(x)}} \quad (4)$$

clipping bounds depend on the old token probability $q(x)$ and the hyperparameters ϵ_{low} and ϵ_{high} , as specified in Equation (4).

The practical effect is that low-probability regions receive more admissible update space, while high-confidence tokens remain controlled by a safe interval. To avoid excessive updates, we follow the motivation of dual clipping (Ye et al., 2020) and cap the maximum bounds for both positive and negative advantages at 10. Appendix A.5 derives the bounds from the importance-sampling perspective, and Appendix A.6 analyzes their behavior.

3 SMOOTH ADVANTAGE STANDARDIZATION(SAS)

The second DCPO component targets response-level utilization by changing how advantages are standardized. GRPO, DAPO, and GSPO compute the advantage $\hat{A}_{new,j}^i$ for the j -th response by standardizing only the rewards $R_{j=1,\dots,G}^i$ generated for the same prompt at step i , as shown by Equation (11) in Appendix A.3. This step-local view creates two sources of lost signal: if sampling randomness makes all rewards identical at one step, all corresponding advantages become zero; and if high-entropy sampling produces skewed label counts, standardized advantages can fluctuate sharply or even reverse sign across steps.

SAS uses the observation that RLVR mainly refines the policy by changing the likelihood of correct samples rather than completely replacing the response distribution (Yue et al., 2025). Under this view, responses to the same prompt across training can be treated as samples from a shared global reward distribution, motivating the *cumulative* standardization in Equation (5).

$$\hat{A}_{total,j}^i = \frac{(R_j^i - \mu_{total}^i)}{\sigma_{total}^i} \quad (5)$$

Here, the statistics are computed over all responses generated so far for the same prompt, so the current update can reuse historical reward information instead of depending only on one noisy step.

To keep the cumulative signal from becoming too rigid while also reducing stepwise noise, SAS introduces two smoothed advantages, $\hat{S}A_{new,j}^i$ and $\hat{S}A_{total,j}^i$. They are weighted averages of step-specific standardization ($\hat{A}_{new,j}^i$) and cumulative standardization ($\hat{A}_{total,j}^i$), with weights that evolve over step i as defined in Equation (6).

$$\hat{S}A_{new,j}^i = \frac{i-1}{i} \hat{A}_{new,j}^i + \frac{1}{i} \hat{A}_{total,j}^i, \quad \hat{S}A_{total,j}^i = \frac{1}{i} \hat{A}_{new,j}^i + \frac{i-1}{i} \hat{A}_{total,j}^i \quad (6)$$

The two smoothed forms serve complementary roles. $\hat{S}A_{new,j}^i$ increasingly emphasizes the current policy outputs as training proceeds, while $\hat{S}A_{total,j}^i$ increasingly emphasizes the cumulative reward distribution and treats the current step as a local adjustment. DCPO then chooses the smoothed advantage with the smaller absolute value, as in Equation (7), so the update retains signal from identical-reward cases while limiting instability from either step-specific or cumulative fluctuations.

$$\hat{A}_j^i = \begin{cases} \hat{S}A_{new,j}^i, & \text{when } |\hat{S}A_{new,j}^i| < |\hat{S}A_{total,j}^i| \\ \hat{S}A_{total,j}^i, & \text{otherwise} \end{cases} \quad (7)$$

As a result, once a prompt enters optimization, later responses for that prompt can continue to contribute even when the current step alone would assign zero advantage. When identical rewards occur within a step, the responses still update the model through the cumulative component $\frac{1}{i} \hat{A}_{total,j}^i$, which is the specific mechanism by which SAS improves response utilization and data efficiency.

3.1 OTM LOSS

The third DCPO component addresses how response advantages are aggregated into the loss. The **Token-Level Mean loss (TLM)** used by DAPO in Equation (13) balances token contributions, but it can distort the intended ordering among responses of different lengths. For example, suppose response A has advantage 1 and length 500, while response B has advantage 0.5 and length 1500 for the same prompt. Under TLM, A receives weight $\frac{500}{500+1500} = 0.25$ and B receives weight $\frac{1500}{500+1500} = 0.75$, yielding total contributions of $1 \times 0.25 = 0.25$ and $0.5 \times 0.75 = 0.375$. Thus the lower-advantage response B exerts more influence solely because it is longer.

The **Sequence-Level Mean loss (SLM)** used by GRPO in Equation (12) has the opposite issue: averaging across responses weakens advantage magnitudes, effectively reducing them by G after advantage calculation. This batch-level averaging can damage the relative structure that standardization has already established across responses, especially at large batch sizes. OTM keeps the contribution structure reviewer-visible: it averages tokens within each response using $|o_i|$, distributes that response’s importance evenly across its tokens, and avoids averaging away the response-level advantage relation across the batch. We call this **Only Token Mean loss (OTM)**, defined in Equation (8).

$$\mathcal{T}_{\text{DCPO}}(\theta) = \sum_{i=1}^G \frac{1}{|o_i|} \sum_{t=1}^{|o_i|} \min \left(r_{i,t}(\theta) \hat{A}_{i,t}, \text{clip}(r_{i,t}(\theta), 1 - \varepsilon_{\text{low}}, 1 + \varepsilon_{\text{high}}) \hat{A}_{i,t} \right) \quad (8)$$

As in DAPO, DCPO does not add a KL-divergence constraint to this loss. The lower and upper clipping bounds in Equation (8) are the dynamic-adaptive bounds introduced in Section 2, linking the OTM aggregation contribution back to the DAC clipping contribution.

4 EXPERIMENTAL SETUP

Models and Datasets The experimental design tests whether the DCPO contributions transfer across model scale and initialization. We train four models—Qwen2.5-Math-1.5B-Instruct, Qwen2.5-3B (common base), Qwen2.5-Math-7B (math base), and Qwen2.5-14B (common base) (Yang et al., 2024a;b)—on a merged corpus of approximately 25k mathematical problems. The corpus combines DAPO-Math-17K (Yu et al., 2025) with levels 3-5 of the MATH subdataset (Hendrycks et al., 2021b; Liu et al., 2025). All training and evaluation problems use the Qwen-Math template described in Section A.8, keeping the prompt format fixed so comparisons reflect optimizer behavior rather than formatting changes.

Evaluation We evaluate on four widely used mathematical reasoning benchmarks: MATH500 (Hendrycks et al., 2021a), AMC23 (Ouyang et al., 2022), AIME24 (Li et al., 2024), and AIME25 (MAA). The metrics are chosen to separate two claimed benefits of DCPO: Avg@1 measures greedy-answer accuracy, while Avg@32 measures the robustness of the learned policy distribution under sampling. Because MATH500 contains 500 problems, we report only Avg@1 for that benchmark; for the smaller AMC23 (40 problems), AIME24 (30 problems), and AIME25 (30 problems), we report both Avg@1 and Avg@32.

- **Avg@1:** Greedy-decoding accuracy, used to measure the model’s single best prediction.
- **Avg@32:** Average accuracy over 32 sampled responses per problem with temperature 1.0 and top_p 1.0, used to measure robustness and stability of the trained policy distribution.

Baseline Settings We compare DCPO against GRPO and DAPO on all four models, and include GSPO on Qwen2.5-Math-7B and Qwen2.5-14B where that baseline is evaluated. Other settings, including clipping thresholds ϵ_* , the codebase, and shared training hyperparameters, are reported in Appendix A.9; this keeps the main comparison focused on the three DCPO design changes rather than auxiliary implementation differences.

Table 1: Main benchmark comparison for the DCPO contributions. Avg@1 reports greedy decoding, Avg@32 reports 32-sample decoding, boldface marks the best value, and Δ reports the change relative to each baseline.

Model	MATH500 (Avg@1)	AMC23 (Avg@1/32)	AIME24 (Avg@1/32)	AIME25 (Avg@1/32)	Average (Avg@1/32)
Qwen2.5-Math-1.5B-Instruct					
base	73.6	57.5/49.4	10.0/10.0	3.3/6.1	36.1/21.8
GRPO	77.2	70.0/68.4	16.7/14.0	20.0/13.5	46.0/32.0
DAPO	76.0	80.0/70.6	20.0/13.5	14.4/12.5	46.5/32.4
DCPO	77.2	75.0/70.8	20.0/15.6	16.7/12.1	47.2/32.8
Δ_{GRPO}	+0.0	+5.0/+2.4	+3.3/+1.6	-3.3/-1.4	+1.2/+0.8
Δ_{DAPO}	+1.2	-5.0/ +0.2	+0.0/+2.1	+2.3/-0.4	+0.7/+0.4
Qwen2.5-3B[Base]					
base	46.4	27.5/7.8	3.3/0.1	3.3/0.7	20.1/2.9
GRPO	69.2	62.5/51.6	10.0/7.5	6.7/4.5	36.3/21.0
DAPO	72.4	57.5/54.0	10.0/8.3	6.7/3.9	36.6/23.1
DCPO	71.2	62.5/55.8	3.3/7.5	10.0/4.7	37.6/22.7
Δ_{GRPO}	+2.0	+0.0/+4.2	-6.7/ +0.0	+3.3/+0.2	+1.3/+1.7
Δ_{DAPO}	-1.2	+5.0/+1.8	-6.7/-0.8	+3.3/+0.8	+1.0/-0.4
Qwen2.5-Math-7B[Base]					
base	50.4	40.0/19.5	13.3/6.0	3.3/1.5	28.4/9.3
GRPO	81.6	77.5/75.9	36.7/32.1	16.7/16.7	53.1/41.6
DAPO	83.0	72.5/ 80.7	36.7/31.6	23.3/14.9	53.9/42.4
GSPO	84.0	80.0/78.8	40.0/34.9	16.7/16.2	55.2/43.3
DCPO	82.5	82.6/79.8	46.7/38.8	16.7/ 17.2	57.1/45.2
Δ_{GRPO}	+0.9	+5.1/+4.9	+10.0/+6.7	+0.0/+0.5	+4.0/+3.6
Δ_{DAPO}	-0.5	+10.1/-0.9	+10.0/+7.2	-6.6/ +2.3	+3.3/+2.8
Δ_{GSPO}	-1.5	+2.6/+1.0	+6.7/+3.9	+0.0/+1.0	+1.9/+1.9
Qwen2.5-14B[Base]					
base	60.8	47.5/16.4	3.3/1.3	3.3/1.1	28.7/6.3
GRPO	81.2	75.0/65.6	13.3/17.6	13.3/10.5	45.7/31.3
DAPO	83.4	87.5/85.1	16.7/16.4	20.0/15.3	51.9/38.9
GSPO	78.6	77.5/75.0	23.3/16.0	16.7/9.9	49.0/33.5
DCPO	84.6	85.0/79.9	20.0/ 18.2	23.3/19.0	53.2/39.0
Δ_{GRPO}	+3.4	+10.0/+14.3	+6.7/+0.6	+10.0/+8.5	+6.5/+7.7
Δ_{DAPO}	+1.2	-2.5/-5.2	+3.3/+1.8	+3.3/+3.7	+1.3/+0.1
Δ_{GSPO}	+6.0	+7.5/4.9	-3.3/ +2.2	+6.6/+10.1	+4.2/+5.5

5 RESULTS AND ANALYSIS

5.1 MAIN RESULTS

Table 1 is the primary evidence that the three DCPO design choices translate into benchmark gains. It compares GRPO, DAPO, GSPO, and DCPO on MATH500, AMC23, AIME24, and AIME25, reporting both greedy decoding (Avg@1) and sampling-based decoding (Avg@32). Across model scales, DCPO is consistently superior or competitive with GRPO, DAPO, and GSPO where available, supporting its use as a general RLVR policy-optimization method rather than a model-specific adjustment.

Overall Superiority of DCPO On Qwen2.5-Math-1.5B-Instruct, DCPO obtains the best average scores (47.2/32.8), improving over GRPO (+1.2/+0.8) and DAPO (+0.7/+0.4). On Qwen2.5-3B, DCPO reaches (37.6/22.7), exceeding GRPO (+1.3/+1.7) and remaining competitive with DAPO, with a positive Avg@1 margin (+1.0) and only a small Avg@32 gap (-0.4). On Qwen2.5-Math-7B, DCPO achieves (57.1/45.2), surpassing GRPO (+4.0/+3.6), DAPO (+3.3/+2.8), and GSPO (+1.9/+1.9). On Qwen2.5-14B, DCPO reaches (53.2/39.0), improving over GRPO (+6.5/+7.7), DAPO (+1.3/+0.1), and GSPO (+4.2/+5.5). These scale-by-scale results make the central contribution visible: DCPO improves reward-aligned policy learning under both deterministic and stochastic decoding.

Robustness on AIME under Sampling (Avg@32) The AIME24 and AIME25 results are especially important because these competition-style benchmarks stress the exploration and robustness claims. Under Avg@32, which evaluates whether the policy can sample correct responses under high-entropy decoding, DCPO shows its largest gains at the 7B and 14B scales: DCPO-7B obtains 38.8 on AIME24-Avg@32, compared with GRPO (32.1), DAPO (31.6), and GSPO (34.9); DCPO-14B obtains 19.0 on AIME25-Avg@32, compared with GRPO (10.5), DAPO (15.3), and GSPO (9.9). These results support the claim that adaptive clipping and smoothed advantages do more than raise average scores: they improve the reliability of correct generations when multiple plausible outputs are sampled.

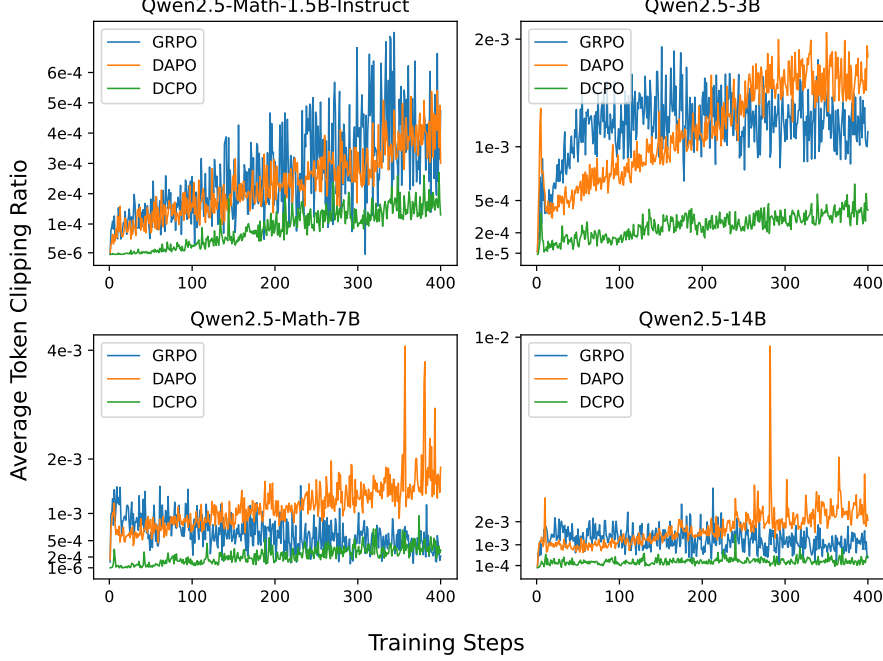


Figure 1: Token Clipping Ratio (TCR) across models and methods, measuring how much token-level learning signal each optimizer discards.

5.2 THE TOKEN CLIPPING RATIO (TCR)

To test the DAC contribution directly, we measure the **Token Clipping Ratio (TCR)**, the proportion of tokens excluded from policy updates during back-propagation because of clipping. Figure 1 shows that the amount of discarded token-level signal differs sharply across reinforcement-learning methods.

$$TCR = Average \left(\sum_{m=1}^N \frac{\text{Number of clipped tokens in } micro_m}{\text{Total number of tokens in } micro_m} \right) \quad (9)$$

where $micro_m$ denotes the m -th microbatch and N is the number of microbatches per training step.

For GRPO, TCR depends strongly on model size: it increases during training for Qwen2.5-Math-1.5B and Qwen2.5-3B, but gradually decreases for Qwen2.5-Math-7B and Qwen2.5-14B. DAPO shows a more uniform upward trajectory, meaning that progressively more tokens are clipped as training proceeds and more updates rely on partial response information. Both GRPO and DAPO also show large fluctuations and occasional spikes, which can destabilize optimization and contribute to entropy collapse. GSPO has an average TCR above 11% on Qwen2.5-Math-7B and above 15% on Qwen2.5-14B, substantially higher than token-level clipping methods and therefore more likely to discard informative token signal.

DCPO exhibits the intended behavior of dynamic-adaptive clipping: across model sizes and training stages, its TCR stays relatively stable and remains an order of magnitude lower than GRPO and DAPO. This lower clipping rate means that DCPO discards fewer tokens, allows more complete

response sequences to affect policy updates, and preserves more exploration space for rare tokens while still controlling outlier updates. The *TCR* result therefore provides mechanism-level evidence for the DAC contribution rather than only an aggregate accuracy comparison.

Qwen2.5-Math-7B illustrates why rare-token clipping matters despite the low average *TCR*. The token distribution quickly concentrates on high-confidence tokens: after roughly 60 steps, about 95% of generated tokens satisfy $(q(x) > 0.9)$, and after roughly 100 steps the fraction rises to about 97% and continues increasing. As shown in Figure 4d, once the old probability satisfies $(q(x) \geq \frac{1}{1+\epsilon_{grpo}} \approx 0.83)$, the new probability $p(x)$ participates in updates within $[\frac{1}{1+\epsilon}, 1]$ without being discarded under any clipping method. Most tokens therefore occupy a never-clipped high-confidence region, explaining the low overall *TCR*. The later-stage *TCR* increases in Figure 1 are consequently concentrated in the low-probability tail, where discarded tokens most directly restrict rare-token exploration.

Prior work (Wang et al., 2025) further supports this interpretation by showing that **high-entropy tokens**, which correspond to low-probability tokens, are important drivers of emergent reasoning ability. Because lower-probability tokens carry more informational content, a broader admissible update range for these tokens expands the effective exploration space available to the model and can support more diverse reasoning behavior.

Together, the empirical *TCR* curves and the high-entropy-token argument justify the DAC design: it relaxes clipping for tokens with $q(x) < \frac{1}{1+\epsilon}$ while preserving the safe interval for high-confidence tokens. By expanding the update interval specifically for the high-entropy tail, DCPO restores learning signal from rare informative tokens and links the clipping contribution to the observed reasoning gains.

5.3 THE RESPONSE UTILIZATION RATIO (*RUR*)

To test the SAS contribution, we measure how often generated responses actually carry gradient signal. The **Response Utilization Ratio** (*RUR*) is the percentage of generated responses with nonzero advantage that participate in the policy update.

$$RUR = \frac{\text{Number of responses with non-zero advantage}}{\text{Total number of generated responses}} \times 100\% \quad (10)$$

Table 2 summarizes the overall average *RUR* for responses generated during GRPO training is low, with an overall mean of only 44.6% across four different models. This indicates that more than half of the generated responses are discarded, highlighting the significant inefficiency in response utilization during the exploration of the diversity. In contrast, our proposed DCPO method achieves an average *RUR* of approximately 70% after the first epoch. This represents a substantial improvement, with an absolute increase of 28% and a relative improvement of 64% over the GRPO’s.

Table 2: Average *RUR* over 400 training steps, measuring response-level gradient utilization.

model	GRPO	GSPO	DCPO
Qwen2.5-Math-1.5B-Instruct	45.6%	-	67.1%
Qwen2.5-3B	48.3%	-	74.3%
Qwen2.5-Math-7B	37.4%	43.5%	73.2%
Qwen2.5-14B	43.9%	47.6%	72.4%
Average	43.8%	45.6%	71.8%

Figure 2 shows the response-level problem that SAS is designed to solve. For GRPO, *RUR* declines substantially over time for every model type and size, often falling from above 90% to below 50%; on Qwen2.5-Math-7B, it drops to about 30% after roughly 200 training steps. This decline indicates that current-step advantage standardization creates many zero-gradient responses and leaves late-stage exploration inefficient. DCPO, by contrast, maintains a high *RUR*: after the first epoch, the ratio stabilizes around 70% and continues to improve with further training.

The *RUR* pattern also clarifies what SAS does and does not try to optimize. It exceeds 90% in the first few steps, then remains around 70% from the second epoch through the end of training rather than approaching 100%. This behavior indicates that SAS recovers useful gradient signal without forcing every generated response to update the model.

The initial *RUR* above 90% is largely caused by early model outputs that fail to follow the required format, receive negative rewards, and inflate the number of nonzero advantages. Once models learn

fixed answer patterns, the more relevant question becomes whether training explores reasoning capability rather than formatting compliance. The later plateau around 70% reflects two remaining zero-advantage cases: (i) many math problems are simple enough that the model solves them consistently, so excluding them from further updates is reasonable because the relevant capability is already stable; and (ii) some problems remain consistently incorrect, either because they are too difficult for the current model even with a correct ground-truth label or because of label noise. In particular, DAPO-Math-17K contains some items whose labels were converted to mislabeled integers, introducing potential "incorrect" labels. Leaving zero-advantage responses unused in these cases acts as a useful filter and reduces the risk of updating on noisy or mislabeled examples.

Because DAPO discards identical-label responses before training, its response use is not directly comparable to GRPO or DCPO under *RUR*. We therefore evaluate DAPO's efficiency by measuring the number of generated responses and the training time required to complete the same number of update steps. To compensate for low sampling efficiency, DAPO uses a generation batch size between 1 and 4 times the training batch size and adjusts it dynamically. Across four models, DAPO requires 3 to 5 times more generated responses than DCPO for the same number of parameter updates, at least doubling training GPU hours without producing better performance. Its utilization also decreases with model size: for Qwen2.5-14B, the proportion of responses used by DAPO falls below 30% after about 300 steps. This comparison highlights the practical contribution of SAS: DCPO improves response utilization without relying on repeated generation.

In summary, the *RUR* and generation-efficiency results show that DCPO improves over GRPO and DAPO by preserving more useful sampled responses, reducing wasted generation, and leaving more room for diversity exploration while improving downstream performance.

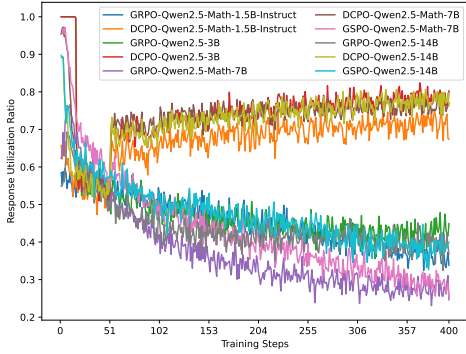


Figure 2: Response Utilization Ratio (*RUR*) during training, showing how much response-level signal remains usable.

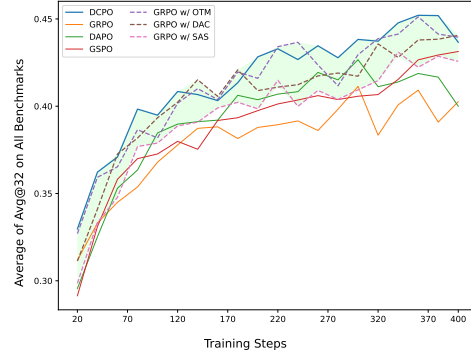


Figure 3: Component ablation on Qwen2.5-Math-7B using average Avg@32, showing the separate and combined effects of OTM, SAS, and DAC.

5.4 ABLATION STUDIES

We use the ablation study on Qwen2.5-Math-7B to isolate the contribution of each DCPO component. The metric is **Avg@32**, because the paper's central claim concerns not only greedy accuracy but also robustness and stability of the sampled policy distribution. For fairness, each ablation modifies one component of the GRPO framework, keeps all other settings fixed, and removes the KL divergence term to align with DAPO, GSPO, and full DCPO. Figure 3 reports learning curves from 20 to 400 steps.

The ablation curves make the contribution structure visible. Each individual component improves inference over the GRPO baseline and reaches or exceeds the performance of DAPO and GSPO in this setting, indicating that OTM, SAS, and DAC each address a distinct bottleneck. Their combination produces the strongest curve: while baseline methods plateau in later training, full DCPO continues to improve, showing cumulative gains in both stability and data-efficient policy learning. The component-level interpretation is as follows.

(i) **Only-Token-Mean loss (OTM)**: Aggregating loss within each response rather than across the whole batch improves over baselines, supporting the claim that preserving response-level advantage relations improves data utilization and reduces fluctuation. (ii) **Smoothed Advantage Standardiza-**

tion (SAS): Replacing traditional stepwise advantage standardization with cumulative smoothing significantly improves GRPO and approaches DAPO, showing that cross-step reward statistics stabilize optimization and improve final performance. (iii) **Dynamic Adaptive Clipping (DAC):** Replacing GRPO’s fixed clipping boundary with probability-adaptive clipping yields substantial gains. DAC alone surpasses GRPO, DAPO, and GSPO in this ablation and is more stable than OTM alone, supporting the value of token-probability-aware exploration. (iv) **Combination of OTM + SAS + DAC (full DCPO):** Combining all three mechanisms achieves the best and most stable results, substantially outperforming GRPO, DAPO, and GSPO. The combined result shows that the components are complementary: OTM preserves response relations, SAS recovers response-level signal, and DAC preserves rare-token exploration.

Overall, the ablation study confirms that each DCPO component contributes positively and that the full method gains from their combination. This directly supports the paper’s contribution claim that improved data efficiency and stable exploration require coordinated changes at the response, advantage, and token-clipping levels.

6 CONCLUSION

This paper contributes **Dynamic Clipping Policy Optimization (DCPO)**, a policy-optimization pipeline for RLVR that makes better use of generated reasoning data at three levels. First, dynamic-adaptive clipping adjusts token-level bounds according to old policy probabilities, expanding exploration for rare low-probability tokens without abandoning clipping control. Second, cumulative smooth advantage standardization reuses reward statistics across optimization steps, reducing the zero-gradient waste caused by current-step standardization. Third, the OTM loss preserves response-level advantage relations by averaging within each response rather than across the whole batch. Experiments on four mathematical reasoning benchmarks and four model scales show that these contributions improve both accuracy and data efficiency: DCPO-7B reaches 38.8 Avg@32 on AIME24, compared with GRPO (32.1) and DAPO (31.6), and DCPO also raises response utilization by 28% over GRPO, doubles training efficiency relative to DAPO, and reduces token clipping by an order of magnitude. These findings support the central conclusion that RLVR benefits when clipping, advantage estimation, and loss aggregation are designed to preserve useful token- and response-level signal. Future work will examine whether the same contribution structure transfers to other domains such as code generation and semantic reasoning.

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A APPENDIX

A.1 MAIN SYMBOL DEFINITIONS

G	Number of responses generated per prompt at each step.
θ	Parameters of the actor model.
$\pi_{\theta}(o_i q)$	Probability of response o_i given prompt q under parameters θ .
$\pi_{\theta_{old}}(o_i q)$	Probability of response o_i given prompt q under previous parameters θ_{old} .
π_{ref}	Reference policy for KL divergence computation.
β	Coefficient for KL divergence term (controls policy update vs. constraint trade-off).
$r_{i,t}(\theta)$	Probability ratio $\frac{\pi_{\theta}(o_i^t q, o_i^{1:t-1})}{\pi_{\theta_{old}}(o_i^t q, o_i^{1:t-1})}$ for token t in response o_i .
$\hat{A}_{i,t}$	Standardized advantage estimate for token t in response o_i .
$ o_i $	The length of the response in tokens.
$\mathbb{D}_{KL}(\pi_{\theta} \pi_{ref})$	KL divergence between current and reference policies.
ε	Probability ratio clipping threshold.
ε_{low} and ε_{high}	Lower/upper thresholds for probability ratio clipping.
M	Maximum allowed response length.
$is_equivalent(a, o_i)$	Binary function indicating whether response o_i matches answer a .
μ_{new}^i	Mean reward across G responses generated at step i (0 if $i = 0$).
σ_{new}^i	Reward variance across G responses at step i (0 if $i = 0$).
μ_{old}^i	Mean reward across $G \cdot (i - 1)$ responses from steps 1 to $i - 1$ (0 if $i \leq 1$).
σ_{old}^i	Reward variance across $G \cdot (i - 1)$ responses from steps 1 to $i - 1$ (0 if $i \leq 1$).
μ_{total}^i	Mean reward across $G \cdot i$ responses from steps 1 to i (0 if $i = 0$).
σ_{total}^i	Reward variance across $G \cdot i$ responses from steps 1 to i (0 if $i = 0$).

A.2 PRELIMINARY

A.3 ADVANTAGE CALCULATION

This appendix first fixes the baseline notation used to identify DCPO’s advantage-standardization contribution. In GRPO and DAPO, the advantage $\hat{A}_{j,t}^i$ for token t in response j is computed by standardizing reward R_j^i against the mean μ^i and standard deviation σ^i of the G responses generated at step i , as shown in Equation (11).

$$\hat{A}_{j,t}^i = \frac{(R_j^i - \mu^i)}{\sigma^i} \quad (11)$$

When all rewards for the same prompt are identical, this baseline rule assigns zero advantage to the responses, so they do not contribute to the model update and become wasted generated data.

A.3.1 GRPO

Shao et al. (2024) proposed **Group Relative Policy Optimization (GRPO)**, which samples G responses for each query, assigns rule-based rewards R , and estimates token-level advantages using Equation (11). The update in Equation (12) is the **Sequence Level Mean loss (SLM)**, which provides the baseline aggregation rule that OTM revises:

$$\mathcal{T}_{\text{GRPO}}(\theta) = \frac{1}{G} \sum_{i=1}^G \frac{1}{|o_i|} \sum_{t=1}^{|o_i|} \min \left(r_{i,t}(\theta) \hat{A}_{i,t}, \text{clip}(r_{i,t}(\theta), 1 - \varepsilon, 1 + \varepsilon) \hat{A}_{i,t} \right) - \beta \mathbb{D}_{\text{KL}}(\pi_{\theta} \parallel \pi_{\text{ref}}) \quad (12)$$

The first term averages token-level advantages within each response and then across all G responses in the batch. The $\mathbb{D}_{\text{KL}}$ penalty constrains the policy against large deviations from a reference policy, such as a pre-trained model, and the fixed clipping threshold ε limits token probability-ratio changes. This widely used formulation is the source of two DCPO motivations: fixed clipping does not adapt to token probability, and batch-level averaging can dilute per-response importance.

A.3.2 DAPO

Yu et al. (2025) proposed **Dynamic sAmpling Policy Optimization (DAPO)** after observing that longer GRPO sequences can exert disproportionate influence on the gradient. DAPO standardizes token counts across the entire batch, producing the **Token Level Mean loss (TLM)** in Equation (13); DCPO keeps this concern visible but addresses it with response-preserving OTM aggregation.

$$\mathcal{T}_{\text{DAPO}}(\theta) = \frac{1}{\sum_{i=1}^G |o_i|} \sum_{i=1}^G \sum_{t=1}^{|o_i|} \min \left(r_{i,t}(\theta) \hat{A}_{i,t}, \text{clip}(r_{i,t}(\theta), 1 - \varepsilon_{\text{low}}, 1 + \varepsilon_{\text{high}}) \hat{A}_{i,t} \right) \quad (13)$$

subject to $0 < |\{o_i \mid \text{is_equivalent}(a, o_i)\}| < G$. This constraint discards prompts whose responses all share the same reward and regenerates responses to maintain batch size, which explains why DAPO improves gradient validity at the cost of substantial data inefficiency.

A.3.3 GSPO

Zheng et al. (2025) proposed **Group Sequence Policy Optimization (GSPO)**, arguing that token-level clipping can inject high-variance noise into gradient estimates. The noise can accumulate over long sequences and be amplified by clipping. GSPO addresses this issue with sequence-wise clipping, providing a useful contrast to DCPO: GSPO changes the clipping level, while DCPO keeps token-level updates but makes their bounds probability-aware.

$$\mathcal{T}_{\text{GSPO}}(\theta) = \frac{1}{G} \sum_{i=1}^G \min \left(s_i(\theta) \hat{A}_i, \text{clip}(s_i(\theta), 1 - \varepsilon, 1 + \varepsilon) \hat{A}_i \right) \quad (14)$$

$$\text{where } s_i(\theta) = \exp \left(\frac{1}{|o_i|} \sum_{t=1}^{|o_i|} \log \left(\frac{\pi_{\theta}(o_{j,t}|q)}{\pi_{\theta_{old}}(o_{j,t}|q)} \right) \right)$$

In our analysis, GSPO discards more than 10% of responses with nonzero advantages, creating response-level data waste. GRPO, by contrast, can retain many high-entropy token-level responses and thereby suffer training instability. These baseline behaviors motivate DCPO's combined design: retain useful responses through SAS and regulate high-entropy tokens through DAC.

A.4 REWARD CALCULATION

The reward function supplies the verifiable supervision signal used by all compared optimizers, so DCPO's gains are attributable to optimization rather than to a different reward model. As shown in Equation (15), each generated response is evaluated for answer correctness and format compliance: +1 is assigned when both are correct, 0 when the format is correct but the answer is incorrect, and -1 when the format is incorrect.

$$R_j^i = \begin{cases} 1, & \text{correct format and correct answer} \\ 0, & \text{correct format but incorrect answer} \\ -1, & \text{incorrect format} \end{cases} \quad (15)$$

Here, j indexes the response group for a given problem at training step i , and the reward depends only on the response format and answer correctness.

A.5 HYPOTHETICAL REASONING PROCESS FOR DYNAMIC-ADAPTIVE CLIPPING BOUNDS

This appendix derives the DAC contribution from the importance-sampling view of policy optimization. Probability-ratio clipping stabilizes reinforcement-learning updates by preventing large parameter shifts, and it is especially important when estimating expectations under a new policy from samples generated by an old policy.

The expected value of a function $f(x)$ under the new probability $p(x)$ can be rewritten as an expectation under the old probability $q(x)$ using an importance-sampling weight, as shown in equation 16:

$$\int f(x)p(x) dx = \int f(x)\frac{p(x)}{q(x)}q(x) dx = \mathbb{E}_{x \sim q} \left[f(x)\frac{p(x)}{q(x)} \right] \quad (16)$$

Although this estimator is unbiased, its variance can be substantially inflated. Equation (17) gives the true variance of $f(x)$ under the new policy, while Equation (18) gives the variance of the importance-sampled estimator under the old policy:

$$\text{Var}_{x \sim p}[f(x)] = \mathbb{E}_{x \sim p}[f(x)^2] - (\mathbb{E}_{x \sim p}[f(x)])^2 \quad (17)$$

$$\begin{aligned} \text{Var}_{x \sim q} \left[f(x)\frac{p(x)}{q(x)} \right] &= \mathbb{E}_{x \sim q} \left[\left(f(x)\frac{p(x)}{q(x)} \right)^2 \right] - \left(\mathbb{E}_{x \sim q} \left[f(x)\frac{p(x)}{q(x)} \right] \right)^2 \\ &= \mathbb{E}_{x \sim p} \left[f(x)^2 \frac{p(x)}{q(x)} \right] - (\mathbb{E}_{x \sim p}[f(x)])^2 \end{aligned} \quad (18)$$

The variance gap in Equation (19) identifies the inflation that clipping is meant to control:

$$\text{Var}_{x \sim q} \left[f(x)\frac{p(x)}{q(x)} \right] - \text{Var}_{x \sim p}[f(x)] = \mathbb{E}_{x \sim p} \left[f(x)^2 \left(\frac{p(x)}{q(x)} - 1 \right) \right] = \int f(x)^2 \left(\frac{p(x)}{q(x)} - 1 \right) p(x) dx \quad (19)$$

PPO Schulman et al. (2017) and RLVR methods (DeepSeek-AI et al., 2024; Liu et al., 2025) mitigate this gap with a fixed symmetric clipping bound on $r(x) = \frac{p(x)}{q(x)}$, defined in Equation (20):

$$|r(x) - 1| \leq \epsilon \quad (20)$$

The limitation is that a fixed clipping bound restricts exploration most severely where the old policy assigns low probability mass. In those regions, the allowable absolute update is small, which restricts meaningful changes to rare tokens. DAPO’s asymmetric clipping partially relaxes the bound, but the constraint remains non-adaptive to the token probability.

Recent work such as Yue et al. (2025) suggests that reasoning paths used by RLVR models are largely already present in the base model’s sampling distribution. The small fraction of clipped parameters during off-policy batch updates also indicates limited parameter shift. This motivates the DAC alternative: replace the fixed bound in Equation (20) with a probability-aware restriction on $r(x)$, as defined in Equation (21).

$$\left| \left(\frac{p(x)}{q(x)} - 1 \right) p(x) \right| \leq \epsilon \quad (21)$$

Substituting $p(x) = r(x)q(x)$ into the restriction yields the expression in Equation (22).

$$-\epsilon \leq (r(x) - 1)r(x) * q(x) \leq \epsilon \quad (22)$$

For the resulting clipping ratio to be valid, the probability variables and their ratio must satisfy standard probability constraints. As shown in Equation (23), $p(x)$ and $q(x)$ are non-negative and at most 1; therefore $r(x)$ and the clipping hyperparameter ϵ must also be non-negative.

$$\begin{cases} 0 \leq p(x) \leq 1 \\ 0 \leq q(x) \leq 1 \\ 0 \leq r(x) \\ 0 \leq \epsilon \end{cases} \quad (23)$$

To make the bound from Equation (22) usable in optimization, we define the necessary bound conditions in Equation (24).

$$\begin{cases} low_{ge}(q(x), \epsilon) = 0.5 - \frac{1}{2}\sqrt{1 + \frac{4\epsilon}{q(x)}} \\ high_{le}(q(x), \epsilon) = 0.5 + \frac{1}{2}\sqrt{1 + \frac{4\epsilon}{q(x)}} \\ low_{le}(q(x), \epsilon) = 0.5 - \frac{1}{2}\sqrt{1 - \frac{4\epsilon}{q(x)}}, & \text{S.t. } q(x)^2 - 4q(x) * \epsilon \geq 0 \\ high_{ge}(q(x), \epsilon) = 0.5 + \frac{1}{2}\sqrt{1 - \frac{4\epsilon}{q(x)}}, & \text{S.t. } q(x)^2 - 4q(x) * \epsilon \geq 0 \end{cases} \quad (24)$$

Combining Equation (22) with Equation (24) gives the following admissible conditions for $r(x)$:

$$\begin{cases} low_{ge}(q(x), \epsilon) \leq 0 \leq r(x) \leq high_{le}(q(x), \epsilon), & \text{when } \text{Var}_{x \sim q} \left[f(x) \frac{p(x)}{q(x)} \right] \geq \text{Var}_{x \sim p} [f(x)] \\ r(x) \leq low_{le}(q(x), \epsilon) \cup high_{ge}(q(x), \epsilon) \leq r(x), & \text{otherwise} \end{cases} \quad (25)$$

For the final clipping interval, we choose the overlapping region centered around ratio 1. This keeps the adaptive rule close to GRPO's conventional fixed bound while minimizing discrepancies between the derived conditions:

$$\begin{aligned} high_{ge}(q(x), \epsilon) \leq r(x) \leq high_{le}(q(x), \epsilon) \\ \text{S.t. } q(x)^2 - 4q(x) * \epsilon \geq 0 \end{aligned} \quad (26)$$

For low_{le} and $high_{ge}$, we ensure the square-root argument is non-negative by taking a maximum with 0, as in $\max(1 - \frac{4\epsilon}{q(x)}, 0)$. This yields the final **dynamic-adaptive clipping bounds for importance sampling**:

$$0.5 + \frac{1}{2}\sqrt{\max\left(1 - \frac{4\epsilon_{low}}{q(x)}, 0\right)} \leq r(x) \leq 0.5 + \frac{1}{2}\sqrt{1 + \frac{4\epsilon_{high}}{q(x)}} \quad (27)$$

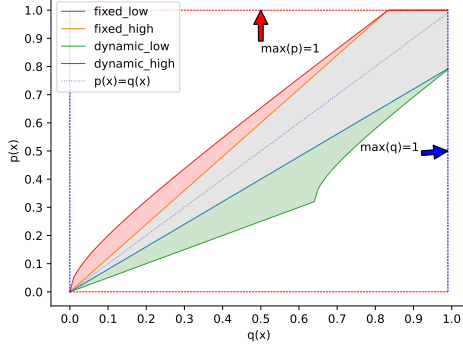
For the upper and lower bounds in Inequality 27, we use separate thresholds ϵ_{high} and ϵ_{low} so the adaptive bound aligns with GRPO at selected reference points while still widening the interval for rare tokens.

A.6 ANALYZING THE DIFFERENCES BETWEEN DYNAMIC-ADAPTIVE CLIPPING AND FIXED CLIPPING

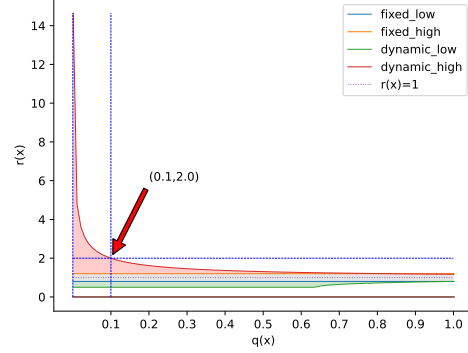
This appendix section explains how the DAC bounds differ from fixed clipping while preserving comparable reference points. To align the dynamic-adaptive bounds with the fixed symmetric bounds at $(\frac{1}{1+\epsilon}, 1)$ and $(1, 1 - \epsilon)$, as shown in Figure 4d, we solve Equation (28) for ϵ_{low} and ϵ_{high} .

$$\begin{cases} q(x) = \frac{1}{1+\epsilon} \\ 1 = 0.5 + \frac{1}{2}\sqrt{1 + \frac{4\epsilon_{high}}{q(x)}} \\ q(x) = 1 \\ 1 - \epsilon = 0.5 + \frac{1}{2}\sqrt{\max(1 - \frac{4\epsilon_{low}}{q(x)}, 0)} \end{cases} \quad (28)$$

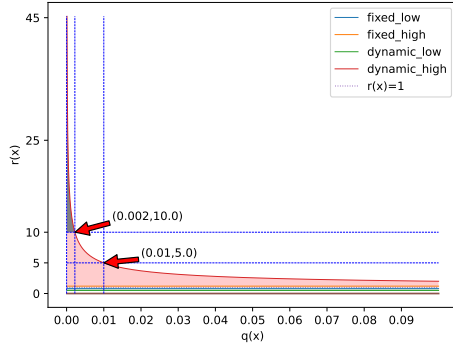
Using the common GRPO-style threshold $\epsilon = 0.2$ gives approximately $\epsilon_{low} = 0.16$ and $\epsilon_{high} = 0.2$. Although these values are asymmetric, their purpose differs from DAPO’s fixed asymmetric clipping: here the asymmetry supports probability-aware alignment and widening rather than a constant upper-bound relaxation.



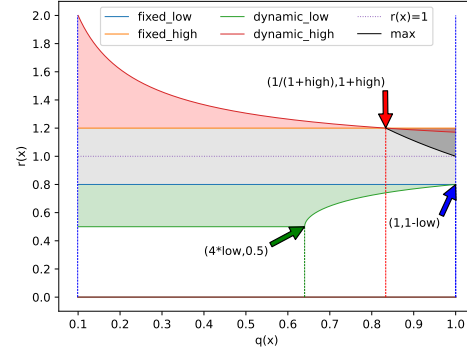
(a) Non-clipped $p(x)$ regions under fixed and adaptive bounds.



(b) $r(x)$ bounds over $q(x) \in [0, 1]$.



(c) $r(x)$ bounds for rare-token probabilities $q(x) \in [0, 0.1]$.



(d) $r(x)$ bounds for moderate-to-high probabilities $q(x) \in [0.1, 1]$.

Figure 4: Fixed versus dynamic-adaptive clipping bounds. Lines compare fixed clipping ($\epsilon = 0.2$) with probability-aware DCPO clipping ($\epsilon_{low} = 0.16$, $\epsilon_{high} = 0.2$).

Dynamic versus fixed clipping: Figure 4 visualizes the key DAC contribution by comparing GRPO’s fixed interval $[1 - \epsilon, 1 + \epsilon]$ with DCPO’s probability-aware interval. For a token with old probability $q(x)$, fixed clipping gives a constant relative width 2ϵ , so the absolute exploration space shrinks linearly as $q(x)$ decreases (Figure 4a). Dynamic-adaptive clipping instead makes the relative width inversely related to $q(x)$, keeping useful exploration space available across the probability range.

Effect of decreasing $q(x)$: When $q(x)$ becomes small, the fixed interval $[1 - \epsilon, 1 + \epsilon]$ remains unchanged and therefore does not increase the model’s ability to update low-probability tokens. The dynamic-adaptive bound expands rapidly in this regime (Figure 4b), creating the larger relative update space that rare-token exploration requires.

Behavior for extremely low probabilities: Figure 4c focuses on $q(x) \in [0, 0.1]$, where the difference is most salient. As $q(x) \rightarrow 0$, the dynamic-adaptive upper bound for $r(x)$ grows rapidly with the inverse square-root form, while the fixed bound remains constant. To prevent unstable importance weights such as gradient explosion or vanishing, we follow the dual-clipping motivation (Ye et al., 2020) and enforce $r_{max} = 10$. The gray region in Figure 4c corresponds to approximately $q(x) \leq 2 \times 10^{-3}$, where the adaptive upper limit is capped at r_{max} .

Upper and lower bounds: For the lower bound, when $q(x) \leq 4\epsilon$, dynamic-adaptive clipping yields $dynamic_{low} = 0.5$, whereas fixed clipping keeps $fixed_{low} = 1 - \epsilon = 0.8$. As $q(x)$ increases beyond

this threshold, the dynamic-adaptive lower bound smoothly contracts toward the fixed value. Similarly, the dynamic-adaptive upper bound exceeds the fixed bound whenever $q(x) < \frac{1}{1+\epsilon_{\text{high}}}$ and merges with it for larger $q(x)$. The adaptive interval $[dynamic_{\text{low}}(q(x), \epsilon_{\text{low}}), dynamic_{\text{high}}(q(x), \epsilon_{\text{high}})]$ therefore widens the admissible region for rare tokens while preserving fixed-bound tightness for common tokens (Figure 4d).

Positive versus negative advantage: The loss in Equation (8) contains both positive- and negative-advantage cases. For positive advantages, the minimum operation clips $r(x)$ within $[0, dynamic_{\text{high}}(q(x), \epsilon_{\text{high}})]$; for negative advantages, the relevant interval is $[dynamic_{\text{low}}(q(x), \epsilon_{\text{low}}), \infty)$. Because excessively large $r(x)$ can destabilize training, we cap the ratio for both positive and negative advantages at $r_{\text{max}} = 10$. This precaution follows Ye et al. (2020) and keeps adaptive clipping from producing excessively high importance weights.

In short, for both positive and negative advantages, dynamic-adaptive clipping gives rare low-probability tokens more exploration space than fixed clipping while retaining an explicit stability cap.

A.7 THE DEFINITIONS OF THE ADVANTAGE STANDARDIZATION

This section provides the statistics behind the SAS contribution. Let i denote the i -th iteration, where prompts generate responses and model parameters are updated; the following quantities define the reward normalization used for current, historical, and cumulative responses.

The response-reward means are defined in Equation (29) for the current step, previous steps, and the cumulative set.

$$\begin{cases} \mu_{\text{new}}^i = \frac{1}{G} \sum_{j=1}^G R_j^i \\ \mu_{\text{old}}^i = \frac{1}{G*(i-1)} \sum_{k=1}^{i-1} \sum_{j=1}^G R_j^k \\ \mu_{\text{total}}^i = \frac{1}{G*i} \sum_{k=1}^i \sum_{j=1}^G R_j^k \\ \quad = \frac{1}{i} (\mu_{\text{new}}^i + (i-1) \mu_{\text{old}}^i) \end{cases} \quad (29)$$

The corresponding response-reward variances are defined in Equation (30), giving SAS the variance terms needed to combine current and cumulative standardization.

$$\begin{cases} \sigma_{\text{new}}^i = \sqrt{\frac{1}{G} \sum_{j=1}^G (R_j^i - \mu_{\text{new}}^i)^2} \\ \sigma_{\text{old}}^i = \sqrt{\frac{1}{G*(i-1)} \sum_{k=1}^{i-1} \sum_{j=1}^G (R_j^k - \mu_{\text{old}}^i)^2} \\ \sigma_{\text{total}}^i = \sqrt{\frac{1}{G*i} \sum_{k=1}^i \sum_{j=1}^G (R_j^k - \mu_{\text{total}}^i)^2} \\ \quad = \sqrt{\frac{1}{i} \left(\sigma_{\text{new}}^i{}^2 + (i-1) \sigma_{\text{old}}^i{}^2 + \frac{i-1}{i} (\mu_{\text{old}}^i - \mu_{\text{new}}^i)^2 \right)} \end{cases} \quad (30)$$

To reduce resource usage when computing σ_{total}^i and μ_{total}^i in Equation (30) and Equation (29), we use equivalent calculations from μ_{new}^i , σ_{new}^i , μ_{old}^i , and σ_{old}^i rather than recomputing over all original rewards.

A.8 THE TEMPLATE FOR EXPERIMENTS

All experiments use the following Qwen-Math-based template (Yang et al., 2024b), ensuring that the reported differences come from the optimization methods rather than prompt-format changes.

Qwen-Math Template

```
<|im_start|>system
Please reason step by step, and put your final answer within \boxed{ }. <|im_end|>
<|im_start|>user
{problem}
<|im_end|>
<|im_start|>assistant
```

A.9 TRAINING SETTING

The training settings keep baseline comparisons aligned while preserving each method’s intended clipping rule. GRPO uses a symmetric clipping threshold $\epsilon = 0.2$; DAPO uses lower/upper clipping thresholds $(0.2, 0.28)$, an additional 512-token soft-punish cache, and maximum probability ratio 10 for negative advantages; GSPO uses lower/upper clipping thresholds $(3e - 4, 4e - 4)$; and DCPO uses low/high clipping thresholds $(0.16, 0.2)$. These DCPO values preserve the same reference clipping points $(q(x), r(x)) = (\frac{1}{1+\epsilon}, 1 + \epsilon)$ and $(q(x), r(x)) = (1, 1 - \epsilon)$ as GRPO (Figure 4d), while setting the maximum probability ratio to 10 for both positive and negative advantages, unlike DAPO.

Training runs for 400 steps with response-generation batch size 512 and parameter-update mini-batch size 32. We use temperature 1 and top_p 1 to generate $G=16$ responses per problem. The prompt maximum length is 1,024 and the response maximum length is 3,072, which is substantially smaller than the 20k maximum length used in the original DAPO.

We adapt the training codebase from Verl (Sheng et al., 2024) and train on 32xH20 GPUs. Unless otherwise stated, hyperparameters match DAPO. For loss computation, we retain the original $\frac{1}{G}$ or $\sum_{i=1}^G \frac{1}{|o_i|}$ weighted-average formulations from GRPO or DAPO rather than Verl’s approximate weighted calculation, ensuring compatibility across different microbatch sizes.

A.10 THE DETAILED PERFORMANCE OF MODELS

Avg@1 Trend Figure 5 reports the average Avg@1 trend by averaging greedy-decoding results on MATH500, AMC23, AIME24, and AIME25. DCPO consistently outperforms GRPO and DAPO across all four models and outperforms GSPO on the two models where GSPO is evaluated. The curve is also more stable, avoiding the large GRPO performance drop and the stronger DAPO fluctuations. This appendix result reinforces the main contribution claim under deterministic decoding: DCPO improves reasoning performance while stabilizing training.

Avg@32 Trend Figure 6 reports Avg@32 on AMC23, AIME24, and AIME25, directly evaluating the sampled-policy robustness that DCPO is designed to improve. GRPO shows substantial fluctuation and occasional collapse; for Qwen2.5-14B, GRPO drops sharply after approximately 170 training steps. Across all evaluated model sizes from 1.5B to 14B, DCPO follows a consistently upward trajectory with limited fluctuation, including fast early growth. Compared with DAPO, DCPO achieves stronger performance while using about half the training time and resources. These results connect the aggregate benchmark gains to the mechanism contributions: dynamic-adaptive clipping supports token-level exploration, and smooth advantage standardization improves response-level and label-level diversity use.

Entropy Trend Figure 7 provides additional evidence for the exploration-stability role of DCPO. GRPO shows a sharp entropy decline after an initial warm-up and then stabilizes in a flat low-entropy region. This entropy collapse reflects rapid loss of policy randomness, reducing sampled-response diversity and limiting later policy optimization.

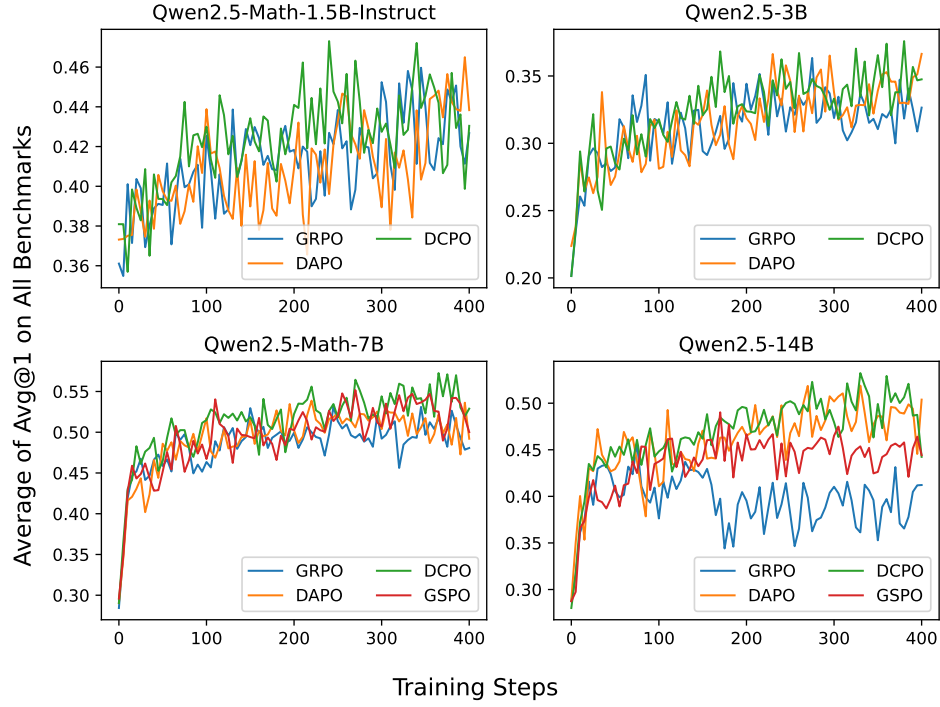


Figure 5: Avg@1 trend across benchmarks, summarizing greedy-decoding gains from DCPO.

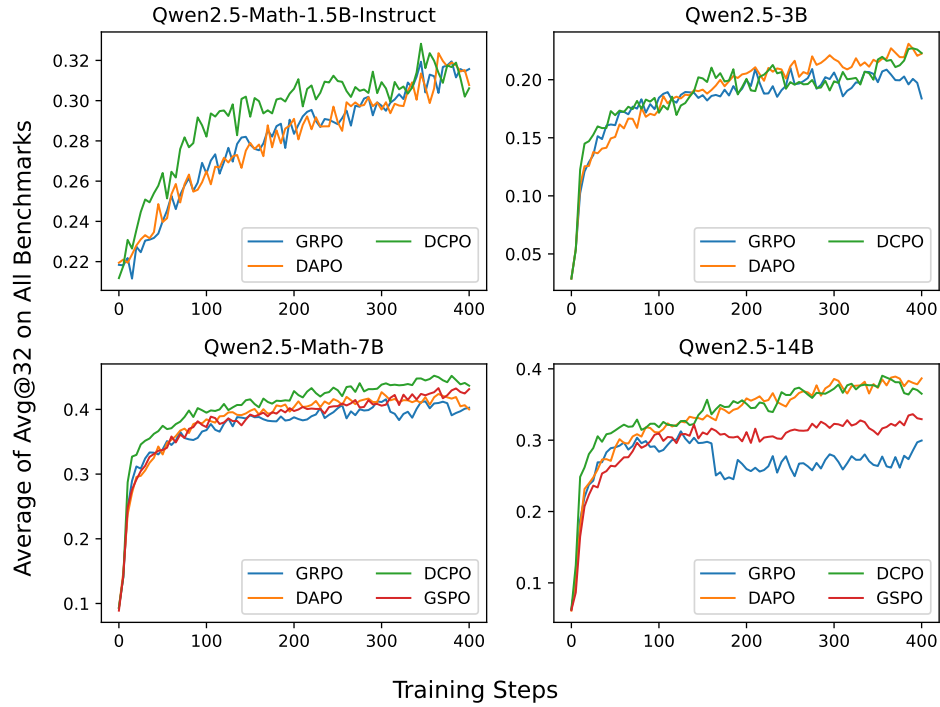


Figure 6: Avg@32 trend across benchmarks, summarizing sampled-policy robustness from DCPO.

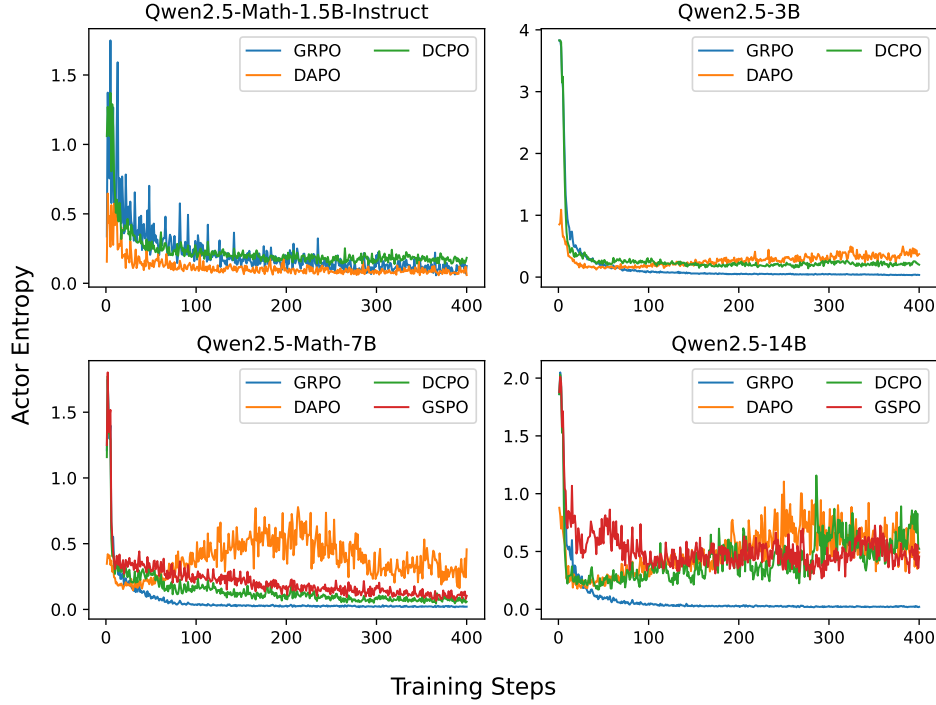


Figure 7: Entropy trends during training, illustrating the exploration-stability balance of DCPO.

The contrast between DAPO and DCPO depends on model type. On the Instruct model (Qwen2.5-Math-1.5B), DAPO, DCPO, and GRPO follow similar declining entropy trajectories. On the larger base models (Qwen2.5-3B, Qwen2.5-Math-7B, and Qwen2.5-14B), the methods separate more clearly: DAPO maintains the highest entropy but with large fluctuations, while DCPO stays in a moderate entropy band. On Qwen2.5-Math-7B, GSPO has entropy similar to DCPO and comparable average performance; on Qwen2.5-14B, GSPO shows a fluctuating downward performance trend. DCPO’s curve is smoother and remains between GRPO’s low-entropy region and DAPO’s high-variance region.

These observations suggest a U-shaped relationship between entropy and training performance. Very low entropy removes exploratory signal and encourages premature convergence, while very high or unstable entropy weakens training stability. DCPO maintains entropy in a moderate range, which is consistent with its intended contribution: preserve enough exploration through adaptive clipping while stabilizing optimization through smoother advantage use. This balance is especially visible as model capacity grows and helps explain DCPO’s stronger inference performance across model sizes.